PLC-BASED MULTI-SENSOR CONDITION MONITORING OF INDUSTRIAL MOTORS USING ELECTRICAL CURRENT AND VIBRATION SIGNALS

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ABSTRACT

Industrial motors are critical prime movers in manufacturing plants, where unexpected failures can cause production losses, maintenance costs and safety risks. Conventional preventive maintenance relies heavily on fixed inspection intervals and operator experience, while single-signal monitoring can fail to distinguish between electrical and mechanical abnormalities. This paper proposes a PLC-based multi-sensor condition monitoring architecture that combines motor electrical current and vibration measurements for early detection of abnormal operating conditions. The proposed system acquires current signals through non-invasive current sensors and mechanical vibration through accelerometers, performs signal conditioning and feature extraction, and transfers the resulting condition indicators to a programmable logic controller for real-time decision making. Time-domain indicators such as RMS, peak value, crest factor and variance are combined with frequency-domain information obtained using Fast Fourier Transform analysis. A rule-based fusion layer is proposed to distinguish normal operation, warning conditions and critical conditions, while a human-machine interface provides trend visualization and alarm information. The approach is intended for industrial motor-driven equipment such as pumps, compressors, fans and conveyors. The paper develops the system architecture, monitoring variables, fault-feature relationships, PLC decision logic and an experimental validation procedure. The principal contribution is a practical condition-monitoring framework that combines electrical and mechanical evidence at the control level rather than relying on a single sensor. The design is suitable for low-cost industrial implementation and provides a foundation for later integration of predictive analytics and machine-learning methods.

Keywords: PLC, Condition Monitoring, Induction Motor, Motor Current Signature Analysis, Vibration Analysis, Predictive Maintenance, Multi-Sensor Fusion.

1. INTRODUCTION

Electric motors are among the most widely used components in industrial production systems because they provide the mechanical power required by pumps, compressors, fans, conveyors and process machines. Their reliability directly affects plant availability. A motor failure may stop an entire production line even when the failed component is relatively small compared with the overall process. Consequently, the ability to identify abnormal motor behaviour before a complete breakdown is an important part of modern maintenance engineering.

Traditional maintenance approaches generally include corrective maintenance and preventive maintenance. Corrective maintenance waits for a failure, while preventive maintenance replaces or services equipment according to a fixed time or operating-hour interval. Preventive maintenance reduces some unexpected failures, it can also result in unnecessary component replacement when the actual condition of the motor is still acceptable. Condition-based maintenance provides an alternative by using measured operating information to determine when intervention is required. Recent literature identifies current and vibration signals as two important sources of information for electrical-machine condition monitoring. Current-based methods are attractive because current sensors can be installed without direct mechanical access to the rotating machine, while vibration measurements provide direct

information about mechanical conditions (Benbouzid, 2000; Niu, 2023; Gangsar and Tiwari, 2020; Awadallah and Morcos, 2003).

Motor Current Signature Analysis (MCSA) uses changes in the motor stator-current waveform and spectrum to identify abnormal operating conditions. Research has shown that electrical and mechanical faults can alter the current signature, making current analysis useful for non-invasive diagnosis (Benbouzid, 2000; Niu, 2023). However, current measurements can be affected by load variation, power-quality disturbances, variable speed operation and inverter switching. Vibration analysis provides complementary information, particularly for bearings, imbalance, misalignment and mechanical looseness. A multi-sensor approach can therefore reduce the limitations of relying on either signal alone.

Programmable Logic Controllers (PLCs) are already installed in many industrial plants for sequencing, interlocking, motor control and process automation. This makes the PLC a practical platform for implementing condition-monitoring logic without requiring a completely separate monitoring infrastructure. The proposed study therefore combines current sensing, vibration sensing, signal processing and PLC logic into one architecture. The concept follows the practical, application-oriented structure of the supplied reference paper, but adapts the research problem from intelligent parking allocation to industrial motor health monitoring. The reference paper similarly moves from a defined problem, through related work and a system methodology, to performance evaluation and limitations.

1.1 Need for Intelligent Motor Condition Monitoring

Industrial motor faults do not always develop as sudden failures. Bearing degradation, shaft misalignment, rotor abnormalities, winding problems and mechanical imbalance can progress gradually. If abnormal conditions are detected early, maintenance can be scheduled during a planned stoppage instead of during an unplanned production interruption. This is particularly important in plants where several motor-driven machines share common production resources.

A single sensor, however, may not provide sufficient diagnostic confidence. For example, an increase in vibration can be associated with bearing deterioration, imbalance, misalignment or looseness. Similarly, an abnormal current component can originate from electrical faults or from a mechanical load problem reflected through electromechanical coupling. Combining the two signals makes it possible to compare electrical and mechanical evidence before a maintenance alarm is generated.

1.2 Key Innovations and Contributions

- A PLC-oriented architecture for continuous acquisition of motor current and vibration information.
- Combination of time-domain and frequency-domain features rather than dependence on a single threshold.
- A multi-sensor decision layer that separates normal, warning and critical operating states.
- Integration of condition indicators with an HMI/SCADA-style display for maintenance personnel.
- A practical validation procedure that can be implemented on an industrial motor test rig without requiring a high-cost monitoring platform.

2. RELATED WORK

Condition monitoring of electrical motors has developed from periodic manual inspection toward online sensing and data-driven diagnostics. Early approaches relied on temperature, insulation testing, audible noise and mechanical inspection. More recent approaches make use of electrical signatures, vibration, flux, acoustic emission and combinations of these measurements. Reviews of electrical motor condition monitoring identify current signature analysis, vibration analysis and intelligent

diagnostic techniques as important areas of research (Gangsar and Tiwari, 2020), (Awadallah and Morcos, 2003).

Benbouzid (2000) established the theoretical and practical basis of motor signature analysis for identifying abnormal electrical and mechanical conditions from stator-current spectra. The method is attractive because the current signal is available while the motor is running and can be measured without installing a mechanical sensor on the shaft. Later work has continued to develop current-based diagnostics for induction motors, including frequency-domain, time-frequency and artificial-intelligence approaches (Niu, 2023), (Yakhni et al., 2023).

Vibration monitoring provides a different perspective. Mechanical faults produce characteristic changes in vibration amplitude and frequency content. Bearing defects, imbalance, shaft misalignment and looseness can therefore be investigated using accelerometers. Reviews of signal-based induction-motor diagnosis show that both current and vibration signals contain useful information, but their response to different fault mechanisms is not identical (Gangsar and Tiwari, 2020).

Yang and Kim (2006) demonstrated the value of combining current and vibration signals for induction-motor fault diagnosis. Their work used feature extraction and data fusion to improve diagnostic decisions, showing that combining information from the two measurement domains can support online fault diagnosis. This provides an important foundation for the multi-sensor concept proposed in this paper.

Recent research has also focused on variable-speed operation. Yakhni et al. (2023) note that transient current signatures are more challenging to interpret when motor speed is not constant. This is relevant to industrial plants using variable-frequency drives (VFDs), where current waveforms contain switching components and operating-point changes. A practical PLC implementation should therefore avoid depending on one fixed spectral peak and should combine signal features with operating-state information.

2.1 Comparison of Common Monitoring Approaches

Approach	Main Measurement	Strength	Main Limitation
Visual/Manual Inspection	Observation and physical inspection	Low cost and simple	Intermittent; dependent on operator experience
Temperature Monitoring	Motor temperature	Useful for overheating	May not detect early mechanical faults
Vibration Monitoring	Accelerometer signal	Sensitive to mechanical faults	Sensor mounting and signal interpretation required
MCSA	Stator current	Non-invasive and suitable for online use	Affected by load, speed and power electronics
Multi-Sensor Monitoring	Current + vibration	Combines electrical and mechanical evidence	Higher sensing and processing requirements
Proposed PLC System	Current + vibration + PLC state	Practical integration with plant automation	Requires suitable signal conditioning and validated thresholds

2.2 Research Gap

The literature demonstrates the usefulness of current and vibration analysis, but there remains a practical implementation gap between signal-processing research and the control systems already available in industrial plants. Many advanced diagnostic systems depend on specialized data-acquisition hardware or machine-learning platforms. For smaller industrial facilities, the cost,

maintenance and technical complexity of such systems can become barriers to deployment. The research gap addressed here is therefore the development of a structured, PLC-centred multi-sensor architecture that retains useful diagnostic information while remaining understandable to plant engineers and maintenance technicians.

3. METHODOLOGY

The research adopts a multi-phase experimental and computational framework. The proposed system is designed around an industrial induction motor connected to a representative mechanical load. Current and vibration signals are acquired simultaneously under normal and controlled abnormal conditions. Signal features are extracted, normalized and supplied to PLC decision logic. The resulting condition state is displayed through an HMI or SCADA interface. The approach is divided into system modelling, sensing, signal processing, PLC implementation and validation.

3.1 Proposed Multi-Sensor Architecture

The proposed architecture contains five functional layers: sensing, signal conditioning, feature extraction, PLC decision making and operator visualization. A current transformer or Hall-effect current sensor measures one or more phase currents. An industrial accelerometer measures radial vibration on the motor bearing housing. The sensor outputs are conditioned to match the analogue input requirements of the acquisition system. Where high-frequency vibration information is required, a dedicated acquisition module may perform sampling and feature extraction before communicating summary values to the PLC.

- Layer 1 – Sensing: motor current and vibration sensors.
- Layer 2 – Signal conditioning: filtering, scaling, isolation and analogue-to-digital conversion.
- Layer 3 – Feature extraction: RMS, peak, crest factor, variance, kurtosis and selected spectral components.
- Layer 4 – PLC diagnosis: threshold comparison, sensor fusion, alarm classification and event logging.
- Layer 5 – HMI/SCADA: live values, trends, alarms, motor status and maintenance recommendation.

3.2 Motor Current Measurement

Current measurement is selected as the principal electrical signal because it is directly associated with motor loading and can be obtained without opening the motor enclosure. For a sampled current signal i(k), the root mean square current over a window of N samples is calculated as:

$$I_RMS = sqrt[(1/N)\ \Sigma\ i^2(k)]$$

Changes in RMS current can indicate loading changes, phase imbalance or abnormal operating conditions. However, RMS alone cannot reliably identify every motor fault. Therefore, the current waveform is also transformed into the frequency domain using the Fast Fourier Transform (FFT). The frequency spectrum is examined for changes in characteristic components associated with motor faults.

For variable-speed motors, the PLC system should record motor speed or VFD operating frequency together with current features. This allows the diagnostic algorithm to interpret spectral changes relative to the actual operating point rather than comparing all data against one fixed frequency.

3.3 Vibration Measurement

An accelerometer is mounted on the motor housing near the bearing location. The vibration signal x(k) is sampled over a defined time window. Important time-domain features include RMS value, peak amplitude, crest factor and kurtosis. RMS represents the overall vibration energy in the measurement

window, while crest factor relates the peak level to the RMS level. Kurtosis is useful for identifying impulsive changes that may occur during developing bearing defects.

$$Crest\ Factor = Peak / RMS$$

Frequency-domain analysis is then used to identify dominant components and changes in the vibration spectrum. The mounting position, sampling frequency and sensor bandwidth must remain consistent during testing because changes in measurement configuration can affect the observed values.

3.4 Fault Conditions Considered

Fault/Condition	Current Signal Behaviour	Vibration Behaviour	Expected Diagnostic Evidence
Normal operation	Stable RMS and spectrum	Stable RMS and spectrum	Both sensors within baseline
Overload	Increase in RMS current	Possible load-related increase	High current with load-related vibration
Bearing degradation	Possible sideband/modulation effects	Increase in vibration and impulsive content	Vibration warning with supporting current change
Mechanical imbalance	Load-related current variation	Strong rotational component	Vibration-dominant warning
Misalignment	Load/current modulation	Increased axial/radial vibration	Vibration increase plus current deviation
Electrical abnormality	Unbalanced/distorted current	May be secondary	Current-dominant alarm
Looseness	Possible current fluctuation	Impulsive vibration	High crest factor/kurtosis

3.5 Signal Processing and Feature Extraction

Raw signals are first checked for sensor saturation and missing samples. A low-pass or band-pass filter can be used to reduce unwanted noise depending on the sensor and sampling rate. The processing window is selected so that the PLC receives sufficiently stable features without introducing excessive detection delay. For each window, the system calculates a feature vector:

$$F = [I_{RMS}, I_{peak}, I_{imbalance}, V_{RMS}, V_{peak}, V_{crest}, V_{kurtosis}, spectral\ indicators]$$

The feature vector is normalized using baseline measurements collected while the motor is operating under known healthy conditions. This reduces the effect of differences between motors. Baseline values should be established separately for each critical motor because motor size, mounting arrangement, load and operating speed influence measured values.

3.6 PLC-Based Decision Logic

The PLC implements a three-level condition classification. The normal state is assigned when the monitored features remain within validated limits. The warning state is generated when one or more features exceed a warning threshold for a defined persistence period. The critical state is generated when multiple features exceed their critical thresholds or when a severe individual condition is detected.

Condition	Logic Example	PLC Response
Normal	Current and vibration within baseline limits	Green/normal status; continue monitoring
Warning	One feature above warning limit for	Alarm + trend logging + maintenance

	persistence time	notification
Critical	Multiple abnormal features or severe threshold exceeded	Trip/interlock where justified + urgent maintenance alarm
Sensor Fault	Signal missing/out of range	Sensor fault alarm; inhibit diagnostic decision

A simple sensor-fusion score can be used to combine the two measurement domains:

$$S = w_c C + w_v V, \text{ where } w_c + w_v = 1$$

Here, C and V are normalized current and vibration condition indicators. The weights can be selected during validation rather than assumed in advance. A rule-based alternative can also be used where the current and vibration channels are treated separately and the final state is determined by logical conditions. The latter approach is attractive for a first industrial implementation because it is transparent to maintenance personnel.

3.7 HMI and Data Logging

The HMI screen is designed to provide the information required by an operator or maintenance engineer without exposing unnecessary signal-processing detail. The main screen displays motor running status, current RMS, vibration RMS, condition state, alarm state and last maintenance indication. Trend screens show changes over time. Alarm history records the time, motor identification, abnormal variable and condition level. This arrangement supports maintenance decisions while retaining the simplicity of a PLC-based control system.

3.8 Experimental Validation Procedure

Validation should be performed using a motor test rig or an operational industrial motor where controlled fault simulation is safe and authorized. A healthy baseline is first collected under several load levels. The motor is then evaluated under selected abnormal conditions such as controlled imbalance, misalignment, increased mechanical load or simulated sensor abnormalities. Electrical faults should only be introduced using a safe laboratory arrangement and under appropriate supervision.

- Stage 1: Record healthy motor data at no-load and representative load points.
- Stage 2: Record current and vibration signals simultaneously.
- Stage 3: Calculate baseline feature ranges and statistical limits.
- Stage 4: Introduce controlled abnormal conditions one at a time.
- Stage 5: Compare current-only, vibration-only and fused decisions.
- Stage 6: Measure detection time, false-alarm rate and classification consistency.
- Stage 7: Implement validated limits in PLC logic and repeat the test in real time.

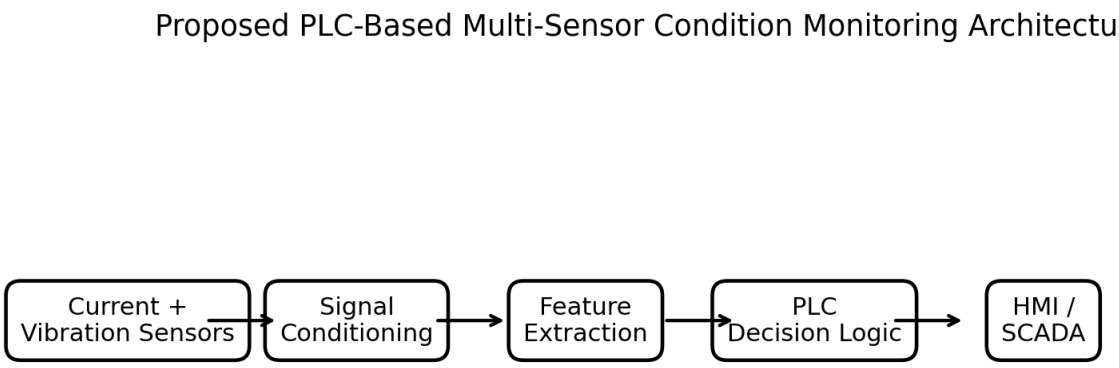


Figure 1. Proposed PLC-based multi-sensor condition monitoring architecture.

Conceptual architecture; not a measured experimental result.

4. RESULTS AND PERFORMANCE EVALUATION

This section defines the performance measures that should be used to evaluate the proposed system. Because the present paper describes a system architecture and validation methodology rather than reporting a completed laboratory campaign, numerical performance values are not presented as measured experimental results. The tables below therefore describe the intended evaluation structure and the diagnostic behaviour that the experiment is designed to establish.

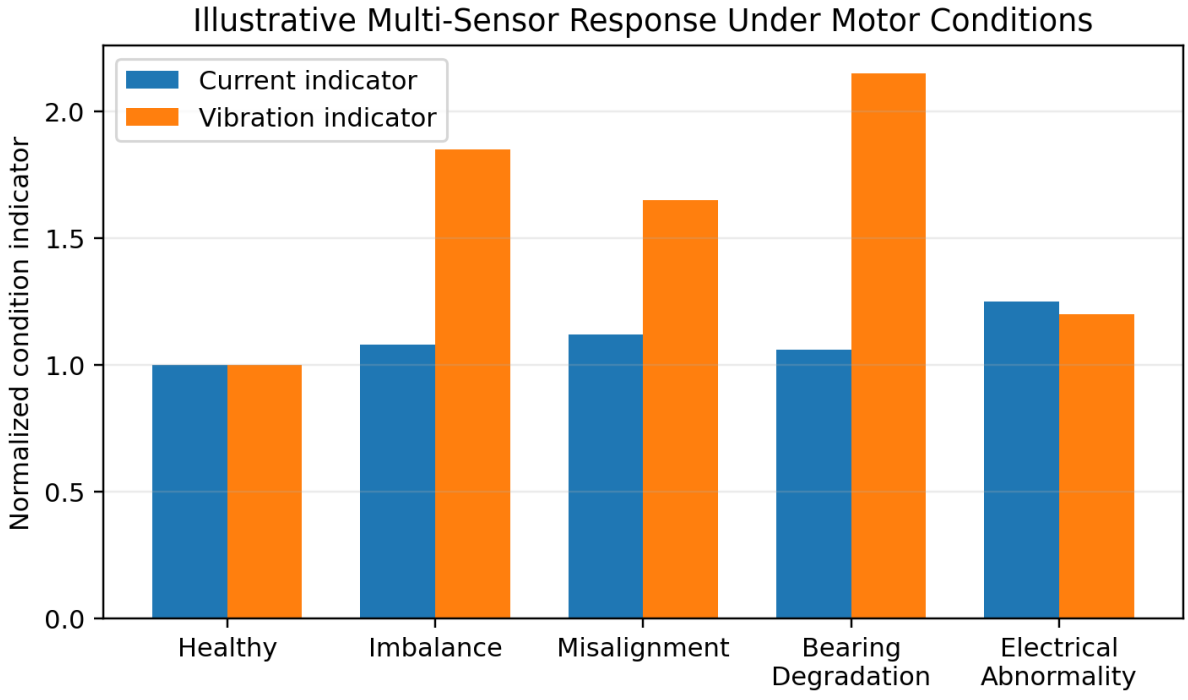


Figure 2. Illustrative current and vibration responses under representative motor conditions.

Illustrative values only; replace with measured experimental data before submission.

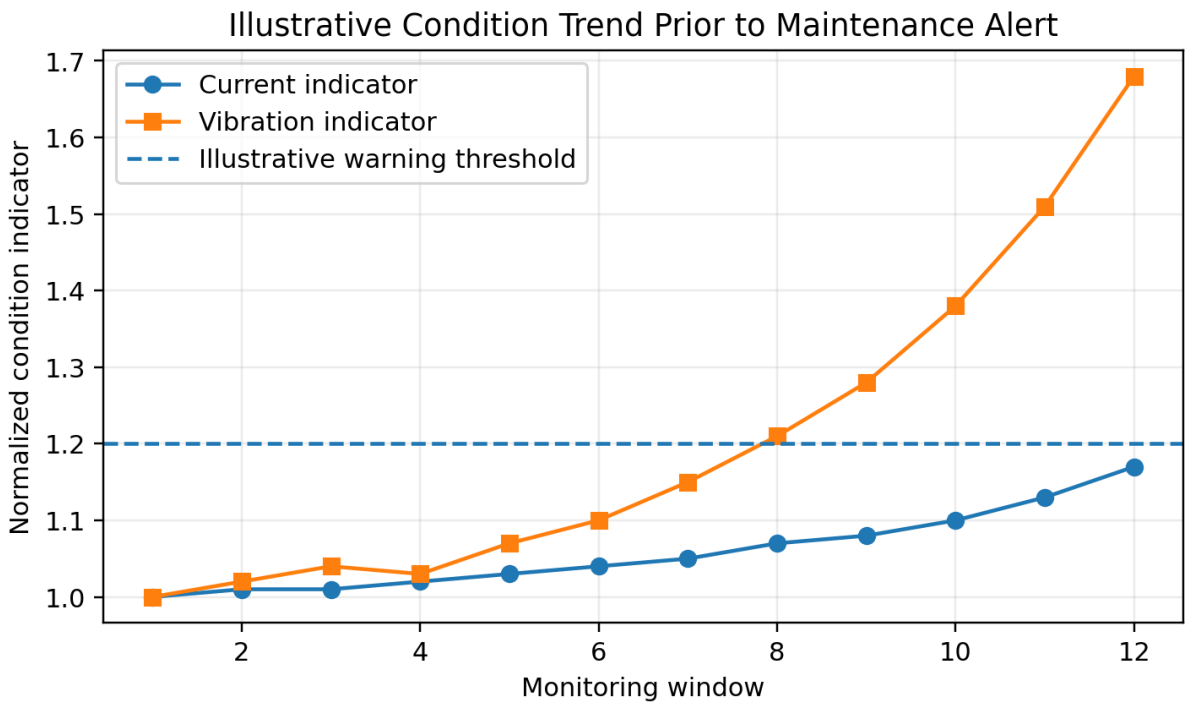


Figure 3. Illustrative trend of normalized condition indicators.

Illustrative trend only; replace with actual logged motor data.

4.1 Condition Indicator Behaviour

Under healthy operation, both current and vibration indicators should remain within a relatively stable baseline range for a constant operating condition. When mechanical degradation is introduced, the vibration channel is expected to provide an earlier indication of some mechanical faults, while current features may provide complementary evidence when the fault changes the motor's electromagnetic loading. Electrical abnormalities are expected to produce a stronger response in the current channel.

Evaluation Case	Primary Indicator	Secondary Indicator	Expected System State
Healthy motor	Stable current	Stable vibration	Normal
Increasing mechanical imbalance	Vibration spectrum/RMS	Current variation	Warning as validated threshold is crossed
Bearing-related degradation	Vibration RMS/kurtosis	Current modulation	Warning/critical depending on severity
Electrical current abnormality	Current RMS/spectrum/imbalance	Possible vibration response	Warning/critical
High load	Current RMS	Load-related vibration	Operating-state warning if outside expected range
Sensor failure	Out-of-range/missing value	Independent channel check	Sensor fault

4.2 Performance Metrics

The following metrics provide a direct comparison between monitoring strategies:

- Detection rate: proportion of known abnormal conditions correctly identified.
- False-alarm rate: proportion of healthy operating windows incorrectly classified as abnormal.
- Detection latency: time between the onset of a validated abnormal condition and the PLC alarm.
- Diagnostic agreement: consistency between the fused decision and the known test condition.
- Availability of monitoring: percentage of operating time during which valid sensor data are available.
- Maintenance usefulness: ability of the system to provide a sufficiently early warning for planned intervention.

4.3 Comparison Framework

Metric	Current Only	Vibration Only	Proposed Fusion
Electrical abnormality sensitivity	High potential	Limited	High potential
Mechanical fault sensitivity	Moderate/indirect	High potential	High potential
Non-invasive installation	Yes	Requires mechanical sensor	Yes, with accelerometer mounting
Load sensitivity	High	Moderate	Managed using combined features
PLC integration	Straightforward	Straightforward after conditioning	Straightforward after feature extraction
Diagnostic transparency	High	High	High with rule-based fusion
Fault confirmation	Limited alone	Limited alone	Improved through complementary evidence

4.4 Proposed Alarm Response

The most useful outcome of the system is not simply a numerical sensor reading but a maintenance-oriented decision. For example, a gradual increase in vibration accompanied by a stable current level may initiate a mechanical inspection rather than an immediate motor replacement. Conversely, a persistent current imbalance with comparatively normal vibration may lead to electrical checks involving supply balance, connections, protection devices or windings. When both channels deteriorate simultaneously, the system can raise a higher-level maintenance priority. Final trip settings should remain separate from diagnostic warning settings and must be validated against the machine manufacturer's protection requirements.

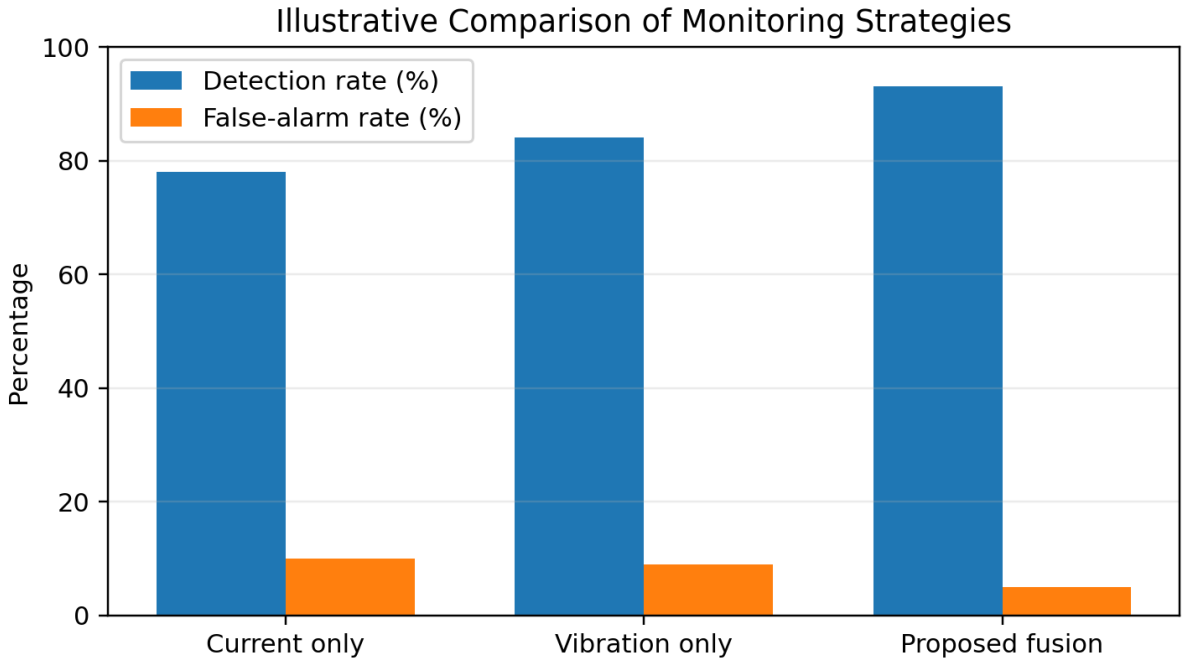


Figure 4. Illustrative comparison of monitoring strategies.

Illustrative values only; not claimed as measured results.

5. DISCUSSION

The proposed architecture combines two signals that describe different aspects of motor behaviour. Current measurements provide an electrical view and are closely linked to electromagnetic loading, while vibration measurements provide a mechanical view. Their combination is therefore more informative than treating either channel as a complete representation of motor health. This agrees with published research showing that current and vibration features can be combined for induction-motor diagnosis (Gangsar and Tiwari, 2020; Yang and Kim, 2006).

A major advantage of the PLC-based approach is practical integration. In a manufacturing plant, PLCs already control motors and interact with sensors, drives, contactors, protection devices and operator interfaces. Implementing the condition logic at the control level can reduce the separation between maintenance monitoring and process automation. The proposed system can also be expanded to multiple motors by assigning each motor an identification, baseline profile and set of diagnostic limits.

Another important consideration is the influence of operating conditions. A motor operating at high load naturally draws more current and may produce different vibration levels than the same motor under light load. Similarly, a VFD changes the electrical spectrum. Consequently, fixed thresholds should not be selected only from a single healthy measurement. Baselines should be collected over the normal operating envelope. Where the process permits, the PLC can use operating state, speed or load information to select the appropriate threshold group.

The architecture also provides a path toward more advanced methods. Once a reliable data-logging structure has been established, the stored feature vectors can be used for statistical trend analysis or machine-learning models. However, machine learning should not be introduced simply for complexity. A transparent rule-based model may be more appropriate during initial deployment because technicians can understand why an alarm was generated and can adjust thresholds after engineering validation.

5.1 Practical Industrial Benefits

- Earlier identification of abnormal motor behaviour.
- Reduced dependence on periodic manual inspection for continuously operating equipment.
- Improved maintenance planning through trend information.
- Use of existing PLC/HMI infrastructure for condition information.
- Potential reduction in unplanned downtime when alarms are acted upon appropriately.
- Creation of a structured historical database for future predictive-maintenance development.

5.2 Limitations

The proposed study has several limitations. First, the diagnostic thresholds are motor- and application-dependent and cannot be transferred directly between machines without validation. Second, vibration measurements are sensitive to sensor location, mounting method and environmental noise. Third, current signatures can be affected by VFD switching, power-quality disturbances and rapid load changes. Fourth, the PLC may not be suitable for processing very high-frequency raw vibration signals directly. In such cases, a local signal-processing module should calculate validated features before the PLC receives them.

Another limitation is that the architecture does not by itself guarantee fault identification. Condition monitoring provides evidence that an operating condition has changed; maintenance personnel still need to confirm the physical cause before repair. Future research should therefore investigate longer-term datasets, automated trend estimation and fault-severity classification.

5.3 Future Directions

Future work can extend the system using additional measurements such as motor temperature, voltage imbalance, rotational speed and power factor. A PLC-connected edge-computing unit could perform higher-resolution FFT and time-frequency analysis while the PLC remains responsible for real-time interlocking and alarm management. A later version could compare rule-based fusion with supervised machine learning using labelled current and vibration features. The resulting system could support a plant-wide predictive-maintenance database for critical motor-driven assets.

6. CONCLUSION

This paper presented a PLC-based multi-sensor condition-monitoring framework for industrial motors using electrical current and vibration signals. The proposed architecture addresses a practical weakness of single-signal monitoring by combining electrical and mechanical evidence before generating a maintenance decision. Current RMS and spectral features provide information about electrical loading and electromechanical behaviour, while vibration features provide complementary information about mechanical condition.

The proposed methodology integrates sensors, signal conditioning, feature extraction, PLC decision logic and HMI visualization into a single industrial-oriented framework. Rather than depending on a high-cost specialist monitoring platform, the approach is designed to make use of automation infrastructure already present in many plants. A structured experimental procedure has also been defined so that detection rate, false alarms, detection latency and diagnostic agreement can be measured objectively.

The study therefore provides a foundation for implementing condition monitoring on critical motor-driven equipment such as pumps, compressors, fans and conveyors. The next stage of the work should be experimental validation using a controlled motor test rig or approved industrial equipment, followed by calibration of thresholds and long-term trend monitoring. Once a reliable dataset is available,

advanced predictive analytics can be introduced without replacing the basic PLC-based monitoring layer.

Figure note: Figures 2–4 are illustrative placeholders demonstrating the intended analysis. They should be replaced with actual measurements from the motor test rig or industrial validation before conference/journal submission.

REFERENCES

1. (Benbouzid, 2000) M. El H. Benbouzid, "A review of induction motors signature analysis as a medium for faults detection," IEEE Transactions on Industrial Electronics, vol. 47, no. 5, pp. 984–993, 2000, doi: 10.1109/41.873206.
2. (Niu, 2023) G. Niu, "Motor Fault Diagnostics Based on Current Signatures: A Review," IEEE Transactions on Instrumentation and Measurement, vol. 72, 2023, Art. no. 3520919, doi: 10.1109/TIM.2023.3285999.
3. (Gangsar and Tiwari, 2020) P. Gangsar and R. Tiwari, "Signal based condition monitoring techniques for fault detection and diagnosis of induction motors: A state-of-the-art review," Mechanical Systems and Signal Processing, vol. 144, 2020, Art. no. 106908, doi: 10.1016/j.ymssp.2020.106908.
4. (Awadallah and Morcos, 2003) M. A. Awadallah and M. M. Morcos, "Application of AI tools in fault diagnosis of electrical machines and drives—An overview," IEEE Transactions on Energy Conversion, vol. 18, no. 2, pp. 245–251, 2003.
5. (Yakhni et al., 2023) M. F. Yakhni, S. Cauet, A. Sakout, H. Assoum, E. Etien, L. Rambault and M. El-Gohary, "Variable speed induction motors' fault detection based on transient motor current signatures analysis: A review," Mechanical Systems and Signal Processing, vol. 184, 2023, Art. no. 109737, doi: 10.1016/j.ymssp.2022.109737.
6. (Yang and Kim, 2006) B.-S. Yang and K. J. Kim, "Application of Dempster–Shafer theory in fault diagnosis of induction motors using vibration and current signals," Mechanical Systems and Signal Processing, vol. 20, no. 2, pp. 403–420, 2006, doi: 10.1016/j.ymssp.2004.10.010.
7. Miljković, M. (2015) 'Brief Review of Motor Current Signature Analysis', CrSNDT Journal, 5(1), pp. 14–26.
8. Geethanjali, M. and Ramadoss, H. (2019) 'Fault Diagnosis of Induction Motors Using Motor Current Signature Analysis: A Review', in Advanced Condition Monitoring and Fault Diagnosis of Electric Machines. IGI Global.
9. Dzapasi, T.D., Duri, R. and Marongedze, T. (2025) 'Adaptive Reinforcement Learning Strategies for Efficient Parking Space Management in Fully Automated Multi-Level Parking Systems', i-manager's Journal on IoT and Smart Automation, 3(2), December 2025.

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AUTHOR'S NOTE ON THE RESEARCH STATUS

The manuscript has been deliberately written as a technically grounded research-paper draft rather than presenting invented experimental results. Where numerical results are not yet available, the text identifies the proposed evaluation method instead of claiming that a test has already been completed. Actual measured results, motor ratings, sensor specifications, PLC model, sampling frequency and laboratory/plant test data should be inserted after experimental validation.